

POSTGRESQL LINUX KERNEL

FRIENDSHIP

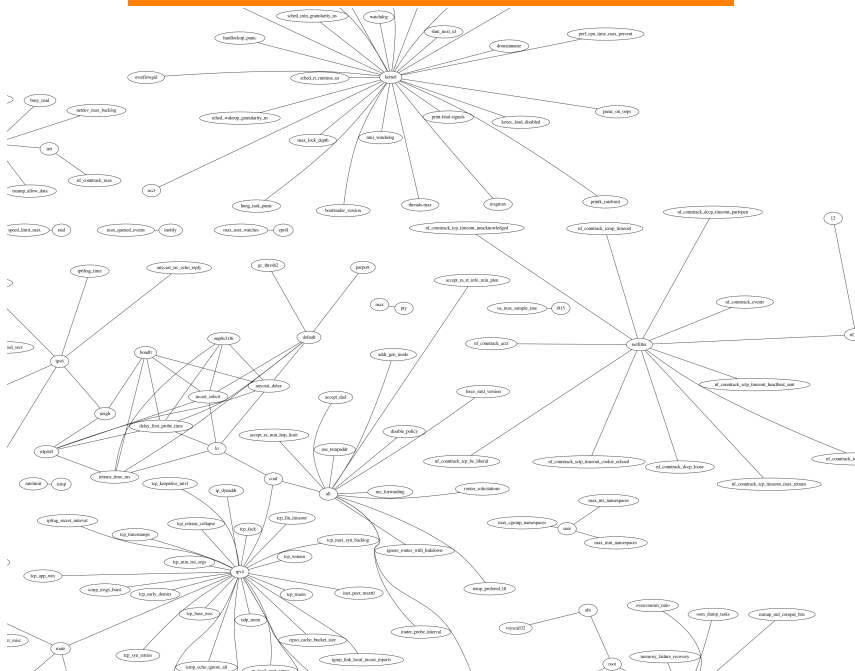
DMITRY DOLGOV

22-10-2018









Plan

Plan

Overview

Plan

Overview

Resources

- Execution scheduling
- Memory management
- Storage IO

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Overview

Resources

- Execution scheduling
- Memory management
- Storage IO

Abstracted layers

- Virtualization
- Containerization

Overview

Bidirectional resource management

IPC mechanisms

New hardware support

...

Execution scheduling

Experiment 1

transaction type: pg_long.sql

latency average = **1312**.903 ms

Experiment 2

SQL script **1**: pg_long.sql

- weight: **1** (targets **50.0%** of total)

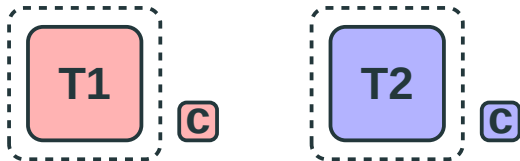
- latency average = **1426**.928 ms

SQL script **2**: pg_short.sql

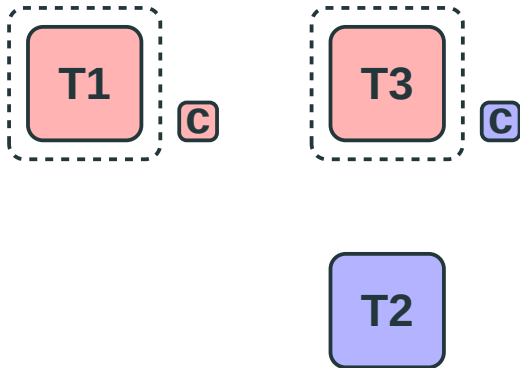
- weight: **1** (targets **50.0%** of total)

- latency average = **303**.092 ms

Scheduling



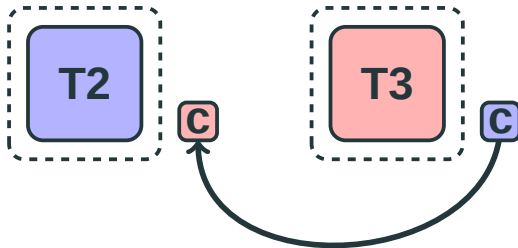
Scheduling



Scheduling



Scheduling



Experiment 1

12,396,382,649

2,750

cache-misses # 28.562%

cpu-migrations

Experiment 2

20,665,817,234

10,460

cache-misses # 28.533%

cpu-migrations

time	cpu	01234	task name [tid/pid]	wait time (msec)	sch delay (msec)	run time (msec)	
4227.834476	[0003]	s	postgres[12935]	0.000	0.000	0.000	
4227.834895	[0003]	s	postgres[12935]	0.000	0.000	0.418	
4227.835478	[0003]	s	postgres[25080]	0.000	0.040	0.583	
4227.836485	[0003]	s	postgres[25080]	0.000	0.000	1.007	
4227.837482	[0003]	s	postgres[25080]	0.000	0.000	0.996	
4227.837784	[0003]	s	postgres[25080]	0.000	0.000	0.302	
4227.837989	[0003]	m	postgres[25080]				migrated: :25077 cpu 1 => 3
4227.837993	[0003]	s	postgres[25080]	0.000	0.000	0.208	
4227.848487	[0003]	s	postgres[25080]	0.000	0.000	10.493	
4227.848991	[0003]	s	postgres[25080]	0.000	0.000	0.504	
4227.849487	[0003]	s	postgres[25080]	0.000	0.000	0.495	
4227.849748	[0003]	s	postgres[25080]	0.000	0.000	0.260	
4227.849912	[0003]	s	postgres[25080]	0.000	0.000	0.164	
4227.851477	[0001]	s	postgres[25082]	0.000	0.000	0.000	
4227.851481	[0002]	s	postgres[25080]	0.000	0.000	0.000	
4227.851778	[0003]	s	postgres[12935]	15.017	0.000	1.866	
4227.852259	[0003]	m	postgres[12935]				migrated: postgres[25083] cpu 1 => 3
4227.852263	[0003]	s	postgres[12935]	0.000	0.000	0.484	
4227.852477	[0003]	s	postgres[25083]	0.000	0.058	0.214	
4227.852478	[0001]	s	postgres[25082]	0.000	0.000	1.001	
4227.852614	[0002]	s	postgres[12935]	0.000	0.000	1.133	

Tunables

Tunables

→ `/proc/sys/kernel/sched_migration_cost_ns`

Tunables

- /proc/sys/kernel/sched_migration_cost_ns
- 500000

Tunables

- `/proc/sys/kernel/sched_migration_cost_ns`
- `500000`
- `/proc/sys/kernel/sched_wakeup_granularity_ns`

Tunables

- `/proc/sys/kernel/sched_migration_cost_ns`
- 500000
- `/proc/sys/kernel/sched_wakeup_granularity_ns`
- 2000000

Tunables

- /proc/sys/kernel/sched_migration_cost_ns
- 500000
- /proc/sys/kernel/sched_wakeup_granularity_ns
- 2000000
- /proc/sys/kernel/sched_min_granularity_ns

Tunables

- /proc/sys/kernel/sched_migration_cost_ns
- 500000
- /proc/sys/kernel/sched_wakeup_granularity_ns
- 2000000
- /proc/sys/kernel/sched_min_granularity_ns
- 1500000

Tunables

- /proc/sys/kernel/sched_migration_cost_ns
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- /proc/sys/kernel/sched_wakeup_granularity_ns
- 2000000
- /proc/sys/kernel/sched_min_granularity_ns
- 1500000
- /proc/sys/kernel/sched_latency_ns

Tunables

- /proc/sys/kernel/sched_migration_cost_ns
- 500000
- /proc/sys/kernel/sched_wakeup_granularity_ns
- 2000000
- /proc/sys/kernel/sched_min_granularity_ns
- 1500000
- /proc/sys/kernel/sched_latency_ns
- 12000000

pgbench and pg_dump

```
real    1m38.990s
user    1m9.127s
sys     0m2.066s
```

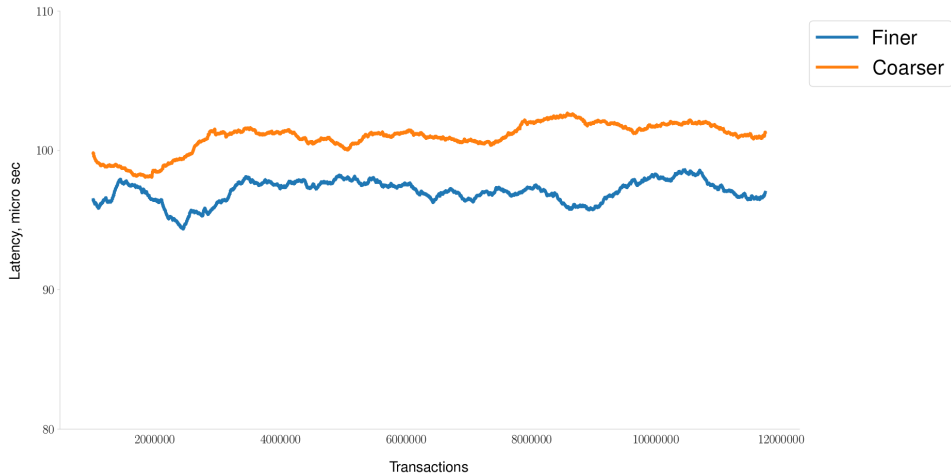
usecs	: count	distribution
0 -> 1	: 16	
2 -> 3	: 4604	**
4 -> 7	: 6812	****
8 -> 15	: 14888	*****
16 -> 31	: 19267	*****
32 -> 63	: 65795	*****
64 -> 127	: 50454	*****
128 -> 255	: 16393	*****
256 -> 511	: 5981	***
512 -> 1023	: 12300	*****
1024 -> 2047	: 48	
2048 -> 4095	: 0	

pgbench and pg_dump

```
real    1m32.030s
user    1m8.559s
sys     0m1.641s
```

usecs	: count	distribution
0 -> 1	: 1	
2 -> 3	: 8	
4 -> 7	: 25	
8 -> 15	: 46	*
16 -> 31	: 189	*****
32 -> 63	: 119	****
64 -> 127	: 96	***
128 -> 255	: 93	***
256 -> 511	: 238	*****
512 -> 1023	: 323	*****
1024 -> 2047	: 1012	*****
2048 -> 4095	: 47	*

Wakeup granularity, microsec



CPU hotplug and HyperThreading

Intel® 64 and IA-32 Architectures Optimization Reference Manual

CPU hotplug and HyperThreading

→ Share execution state and cache

Intel® 64 and IA-32 Architectures Optimization Reference Manual

CPU hotplug and HyperThreading

- Share execution state and cache
- Spin locks have significant impact

Intel® 64 and IA-32 Architectures Optimization Reference Manual

CPU hotplug and HyperThreading

- Share execution state and cache
- Spin locks have significant impact
- PAUSE instruction (skylake latency 140 cycles)

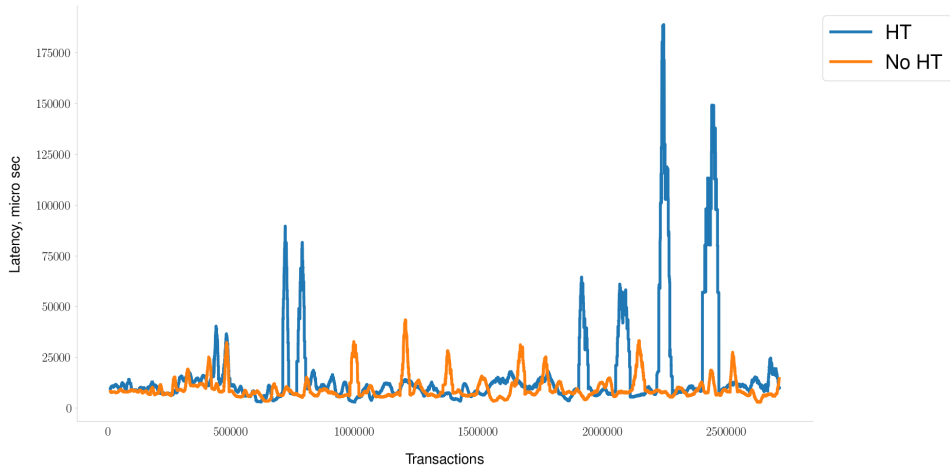
Intel® 64 and IA-32 Architectures Optimization Reference Manual

CPU hotplug and HyperThreading

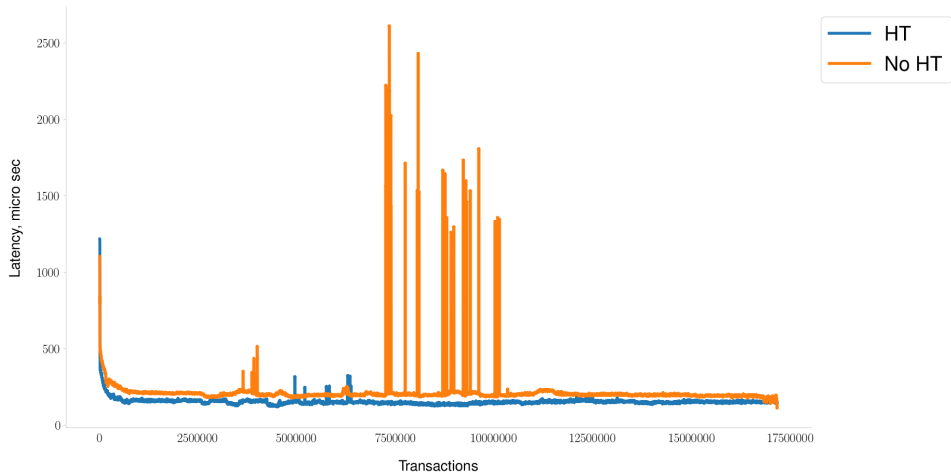
- Share execution state and cache
- Spin locks have significant impact
- PAUSE instruction (skylake latency 140 cycles)
- More deviation for latency

Intel® 64 and IA-32 Architectures Optimization Reference Manual

Latency rolling standard deviation, r/w

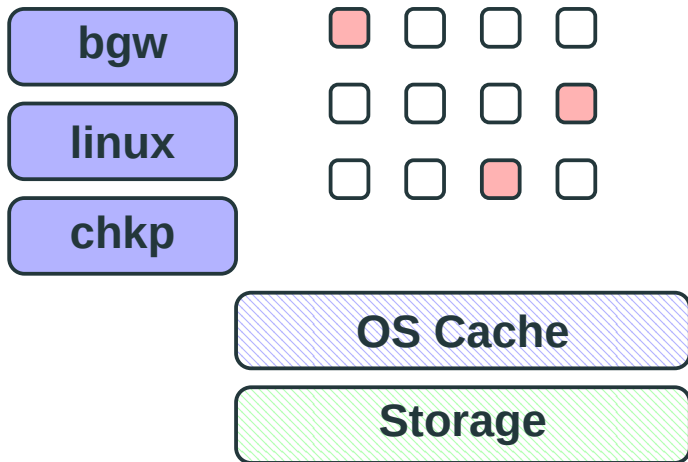


Latency rolling standard deviation, readonly

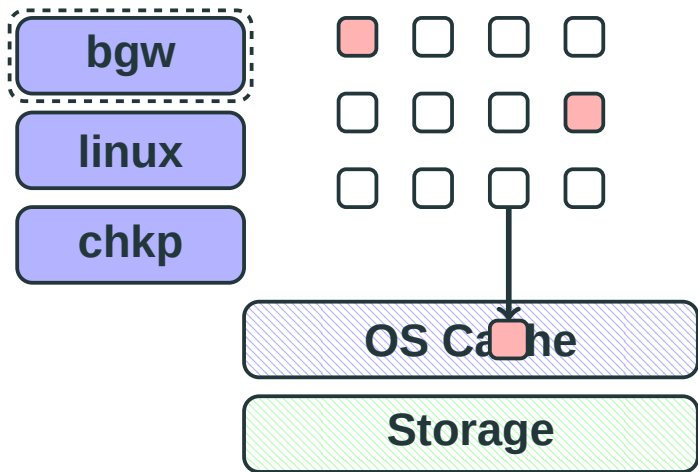


Memory management

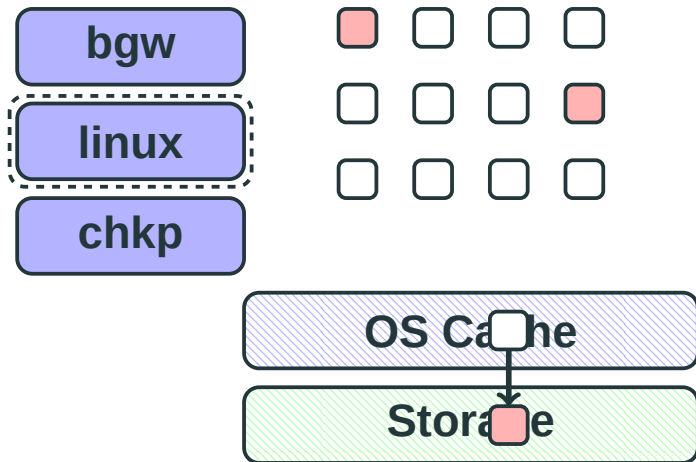
Dirty pages



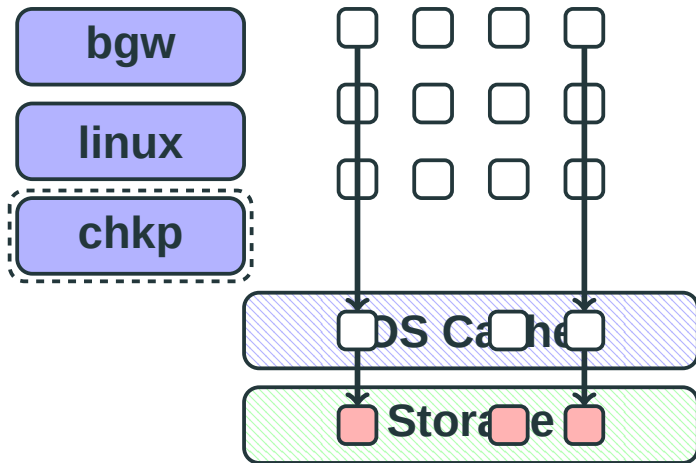
Dirty pages



Dirty pages



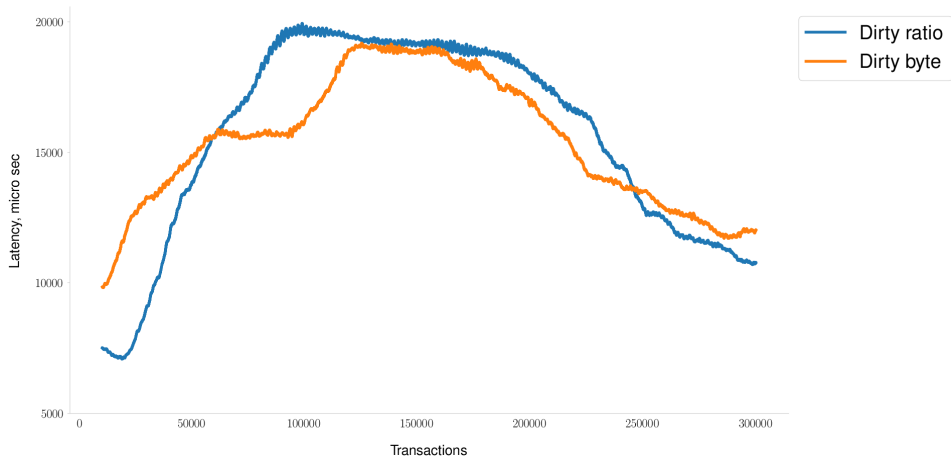
Dirty pages



Dirty pages, r/w

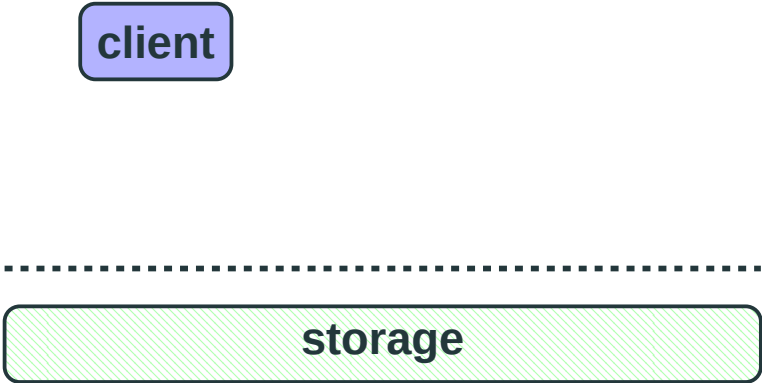
- `vm.dirty_ratio` 20
- `vm.dirty_background_ratio` 10
- `vm.dirty_bytes` 0
- `vm.dirty_background_bytes` 0

Dirty pages



Storage IO

WAL

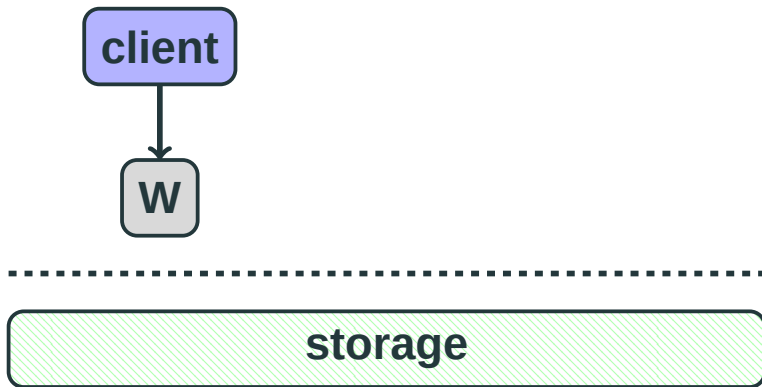


client

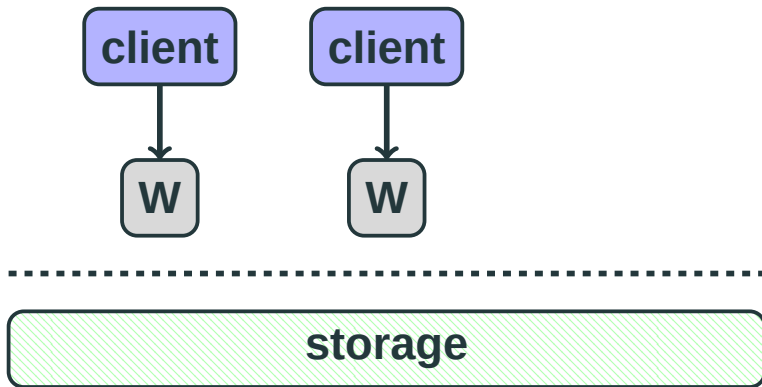
The diagram illustrates a system architecture. At the top, there is a solid orange horizontal bar. Below it, the text 'WAL' is centered. Further down, a light purple rounded rectangle labeled 'client' is positioned. A horizontal dashed line separates the client from the storage component below. The storage component is a light green rounded rectangle with diagonal hatching, labeled 'storage'.

storage

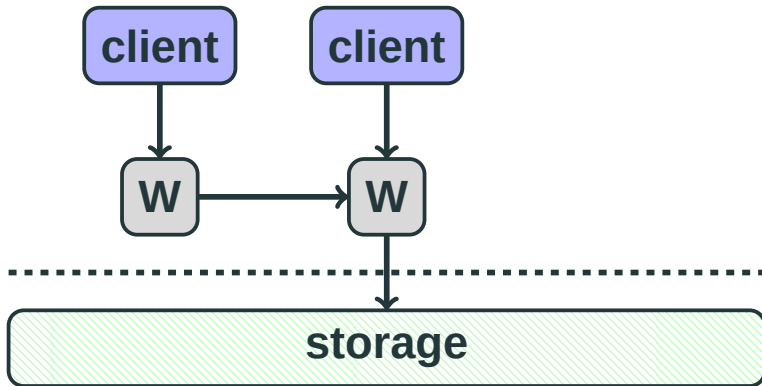
WAL



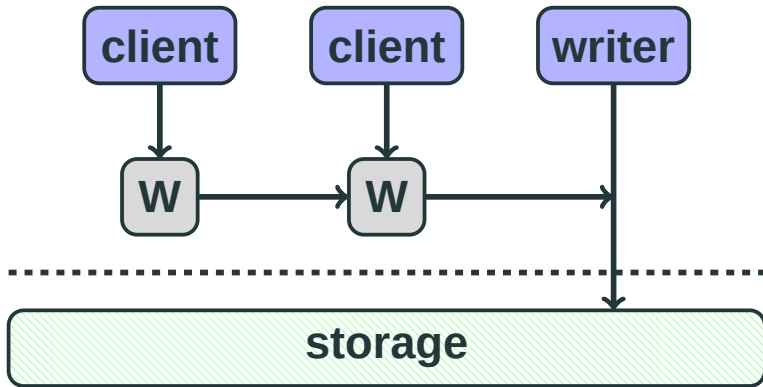
WAL



WAL



WAL



WAL

- Bufferer IO
- fdatasync
- Writeback error propagation

NVMe

- better for resource sharing (PCI express) under the virtualization
- `/sys/block/sda/queue/scheduler` [noop|none]
- DSM operations

NVMe DSM

- Expected lifetime
- Prepare for some workload (read/write)
- Access frequency

NVM Express Revision 1.3c May 24, 2018

DSM support

- Command DWORD 11 in ioctl
- fcntl SET_FILE_RW_HINT
- nvme-cli (ioctl)
- Specify a start block and a range length

```
# get a start block
```

```
hdparm --fibmap data_file
```

```
data_file:
```

```
filesystem blocksize 4096, begins at LBA 0;
```

```
assuming 512 byte sectors.
```

byte_offset	begin_LBA	end_LBA	sectors
0	55041560	55041567	8

```
# set dsm for sequential read optimized
```

```
nvme dsm /dev/nvme1n01 --slbs=55041560 --blocks=1 --idr
```

Virtualization

Timekeeping

Timekeeping in VMware Virtual: Information Guide

Timekeeping

- Statistical sampling
(occasional incorrect charging)

Timekeeping in VMware Virtual: Information Guide

Timekeeping

- Statistical sampling
(occasional incorrect charging)
- Exact measurement (TSC time drift)

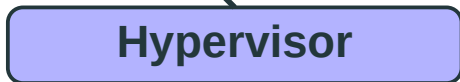
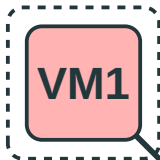
Timekeeping in VMware Virtual: Information Guide

Timekeeping

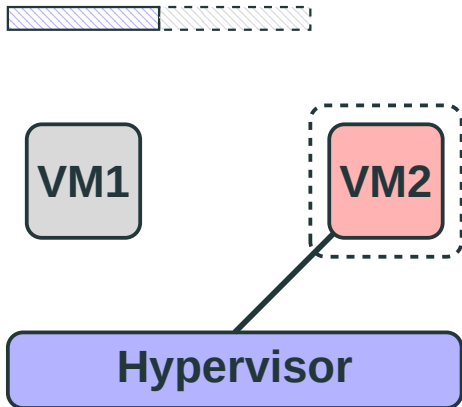
- Statistical sampling
(occasional incorrect charging)
- Exact measurement (TSC time drift)
- `/sys/devices/system/clocksource/clocksource0/`

Timekeeping in VMware Virtual: Information Guide

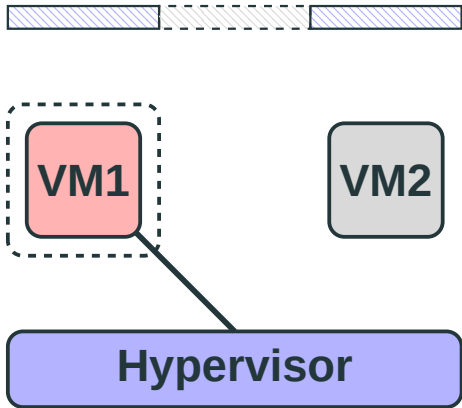
Scheduling



Scheduling



Scheduling

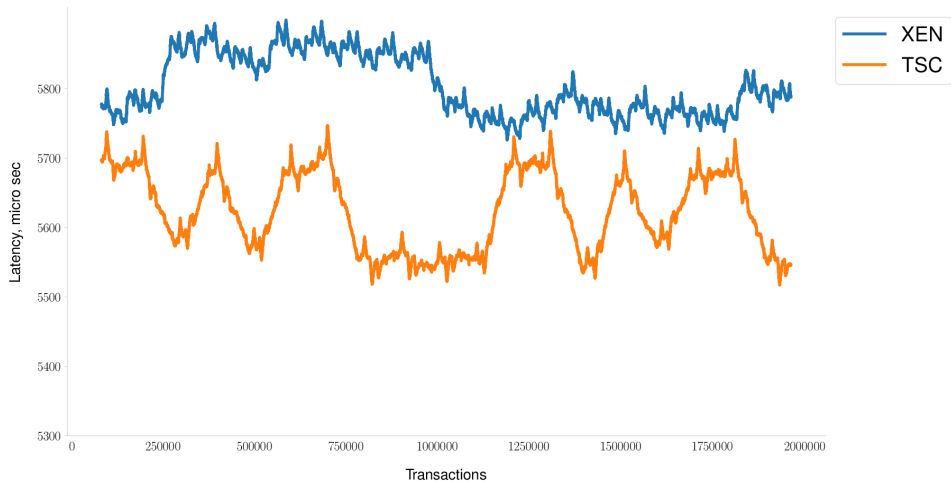


vDSO

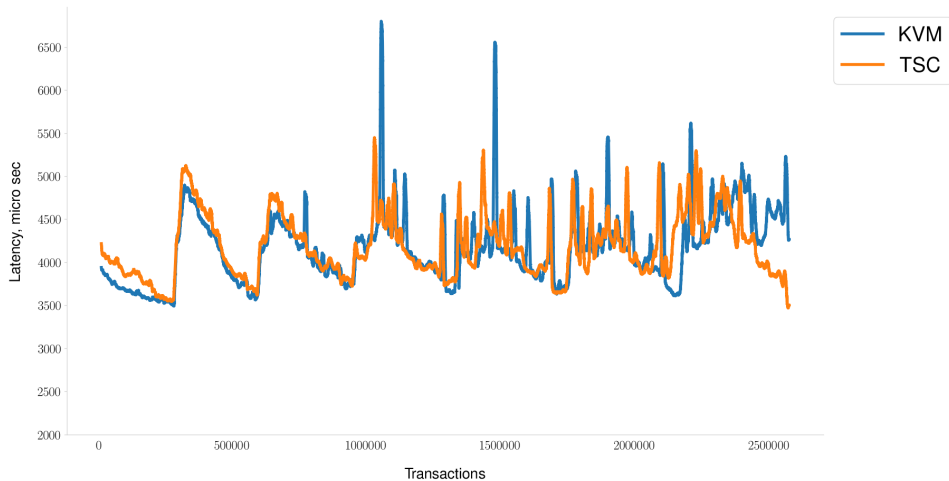
- gettimeofday
- clock_gettime
- XEN doesn't support vDSO for them
- unnecessary context switches to a kernel

Two frequently used system calls are 77% slower on AWS EC2

Latency m4.xlarge XEN/TSC, r/w



Latency m5.xlarge KVM/TSC, r/w



Locks

Intel® 64 and IA-32 Architectures Software Developer's Manual, Vol. 3

Locks

→ Lock holder preemption problem

Intel® 64 and IA-32 Architectures Software Developer's Manual, Vol. 3

Locks

- Lock holder preemption problem
- Lock waiter preemption problem

Intel® 64 and IA-32 Architectures Software Developer's Manual, Vol. 3

Locks

- Lock holder preemption problem
- Lock waiter preemption problem
- Intel PLE (pause loop exiting)

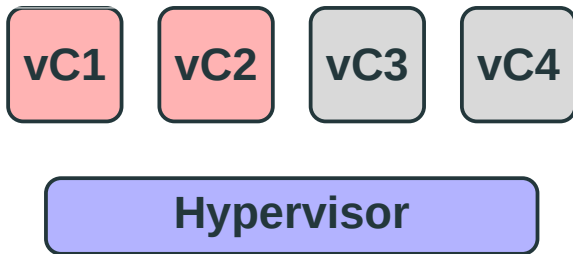
Intel® 64 and IA-32 Architectures Software Developer's Manual, Vol. 3

Locks

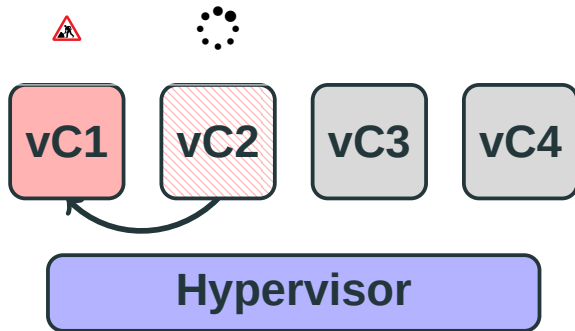
- Lock holder preemption problem
- Lock waiter preemption problem
- Intel PLE (pause loop exiting)
- PLE_Gap, PLE_Window

Intel® 64 and IA-32 Architectures Software Developer's Manual, Vol. 3

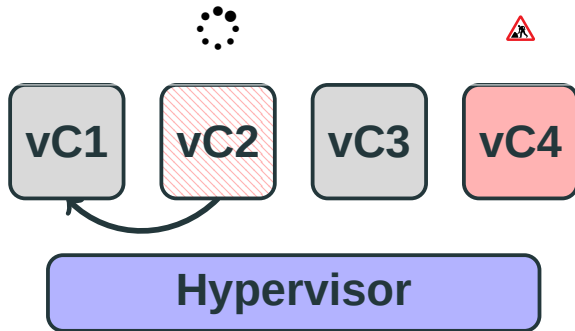
vCPU



vCPU



vCPU



Containerization

cgroups controllers

- cpu,cpuacct
- cpuset
- memory
- devices
- freezer
- net_cls
- rdma

- blkio
- perf_event
- net_prio
- hugetlb
- pids
- rdma

8388 **8388** postgres blk_throtl_bio

blk_throtl_bio+0x1	[kernel]
dm_make_request+0x80	[kernel]
generic_make_request+0xf6	[kernel]
submit_bio+0x7d	[kernel]
blkdev_issue_flush+0x68	[kernel]
ext4_sync_file+0x310	[kernel]
vfs_fsync_range+0x4b	[kernel]
do_fsync+0x3d	[kernel]
sys_fdatasync+0x13	[kernel]
fdatasync+0x10	[libc-2.24.so]
XLogBackgroundFlush+0x17e	[postgres]
WalWriterMain+0x1cb	[postgres]
PostmasterMain+0xfea	[postgres]

bklcio controller

- CFQ & throttling policy (generic block layer)
- No weight related options will work without CFQ
- Advisable io scheduler for SSD is noop/none
- Block layer do sampling to enforce throttling

throttle_sample_time

This is the time window that blk-throttle samples data, in millisecond. blk-throttle makes decision based on the samplings. Lower time means cgroups have more smooth throughput, but higher CPU overhead. This exists only when `CONFIG_BLK_DEV_THROTTLING_LOW` is enabled.

blkio

On traditional cgroup hierarchies, relationships between different controllers cannot be established making it impossible for writeback to operate accounting for cgroup resource restrictions and all writeback IOs are attributed to the root cgroup.

<https://git.kernel.org/pub/scm/linux/kernel/git/torvalds/linux.git>

Bad neighbour

- memory fragmentation
- buddy allocator can fail
to find a page of proper size
- kernel will start a compaction process

Bad neighbour

- PGSemaphore* functions make use of futex
- Per-cpu hash table for futex with hash buckets

Bad neighbour

- WAL segment/heap file creation
- inode lock contention

Understanding Manycore Scalability of File Systems

Questions?

 github.com/erthalion

 github.com/erthalion/ansible-ycsb

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 [9erthalion6 at gmail dot com](mailto:9erthalion6@gmail.com)